fp223

Emergence of Social Reasoning through Adaptive Evolution of LLMs

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How could Theory of Mind arise through evolution? We evolve LLMs as model organisms and find: environment-specific adaptation / an asymmetric knowledge–reasoning trade-off / a gradual shift of internal representations toward ToM.

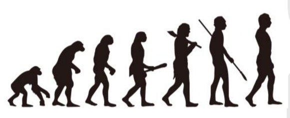
Question

- **The ToM Debate in LLMs is intensifying.**
  Do LLMs have social cognitive abilities (Theory of Mind)? [Kosinski, 2024; Ullman, 2023]
- **Limitations of Existing Studies:**
  benchmark finished models / probe their activations
  → both examine abilities already acquired through training
- **The Missing Piece:**
  How could such abilities arise through evolution?

Purpose

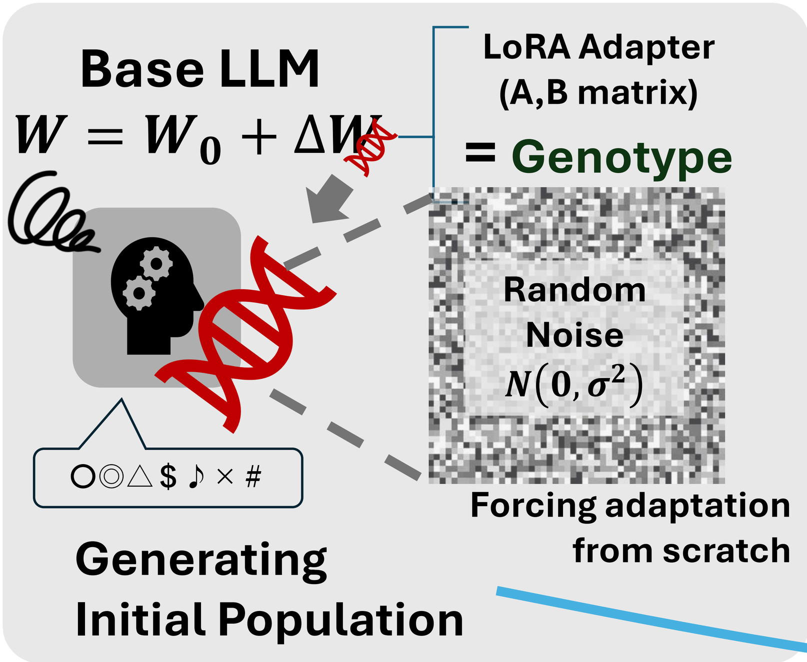
Understand the evolutionary origins of social cognition, using LLMs

- **Theory — Social Brain Hypothesis [Dunbar, 1998]:** hominoid social intelligence arose through adaptive evolution to complex social environments

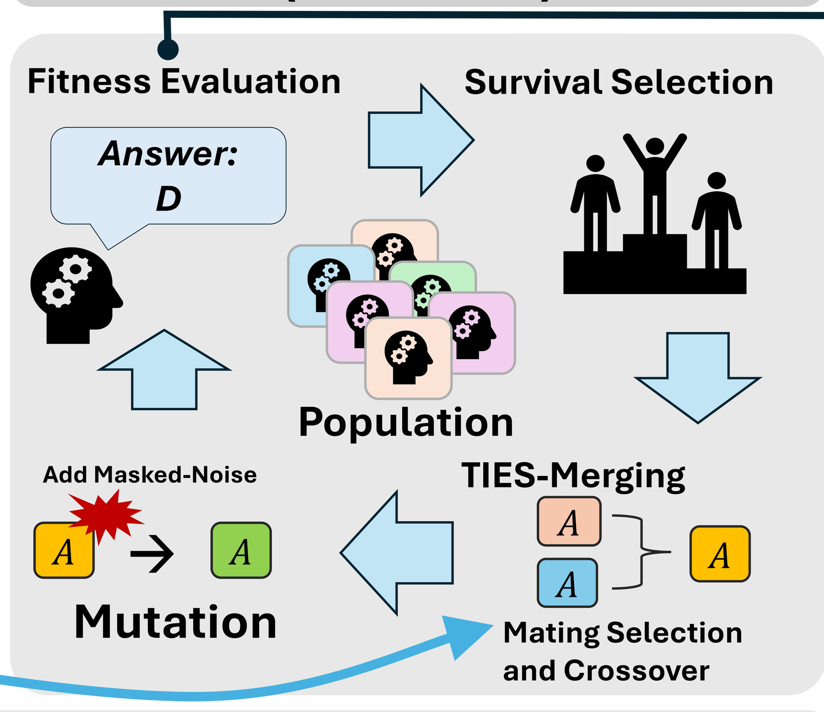


- **Paradigm — LLMs as Computational Model Organisms [Kolling et al., 2025]:** LLMs serve as a controllable platform to simulate and analyze the emergence of cognitive functions.

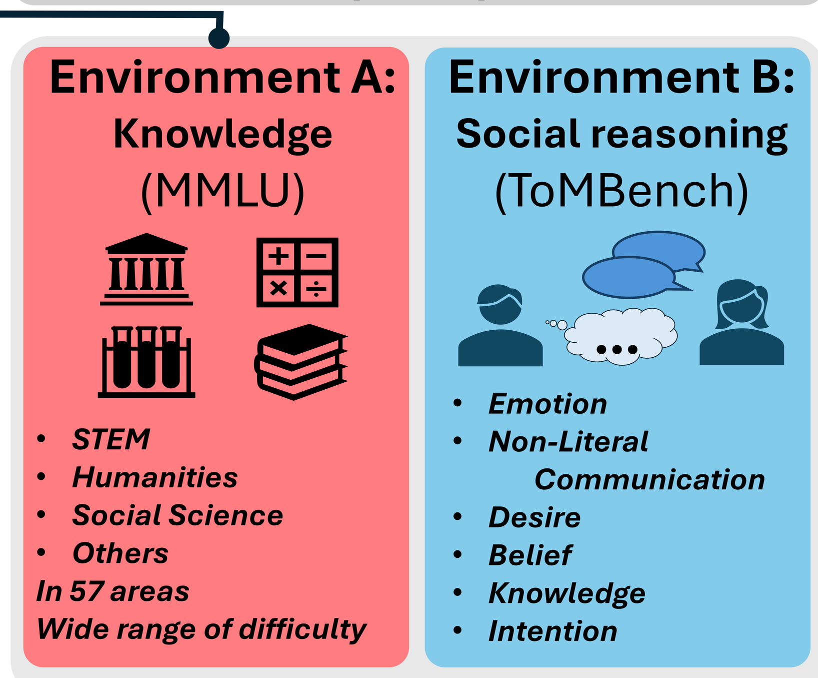
Initial Population (Start adaptation from zero)



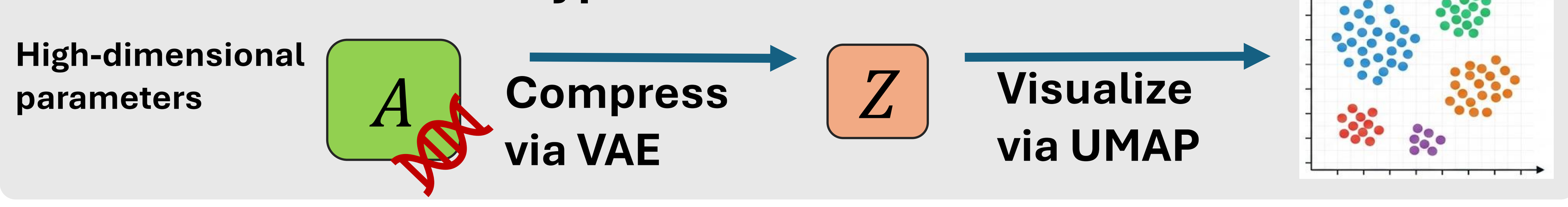
Evolution cycle (GENOME)



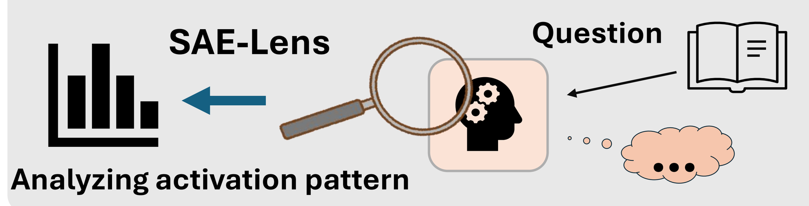
Adaptation environment (Task)



Visualization of Genotypes:



Internal state analysis:



Approach — experimental evolution of LLMs as "social model organism"

Genotype: randomized LoRA adapter
→ behavior degraded to near-random, adaptation starts from scratch

Environment: benchmark tasks as selection pressure
(Knowledge: MMLU / Social reasoning: ToMBench)

Fitness: accuracy in each environment

Method — evolution of LoRA adapters using GENOME [Zhang et al., 2025]

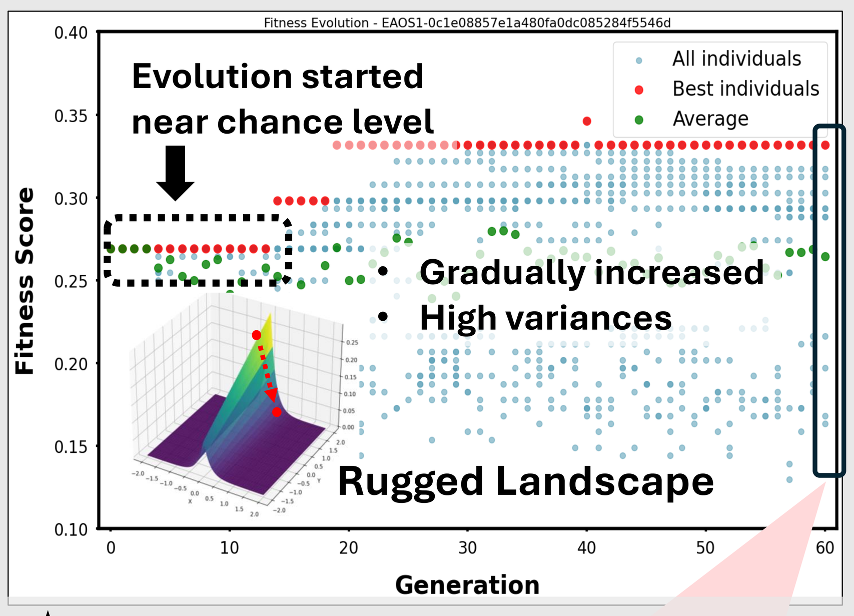
- **Genotype:** LoRA ($W = W_0 + BA$), rank r — only A, B evolve
- **Initialization:** A and $B \sim \mathcal{N}(0, \sigma^2)$ — non-zero B distorts W_0 → near-random start, knowledge retained
- **Evolution:** TIES-Merging crossover / masked Gaussian mutation / tournament selection
- **Fitness:** accuracy on MMLU or ToMBench (chance = 0.25)

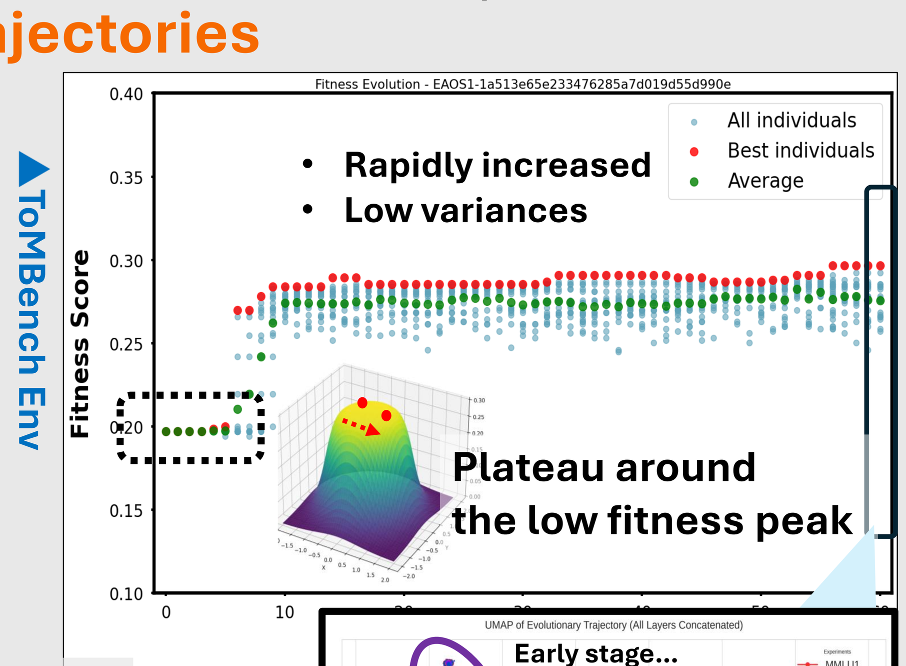
Experiment 1: evolution of fitness & genotypes, generalization across the two environments

LLM	Gemma2-2b-it
Knowledge task	MMLU (Massive Multitask Language Understanding) [Hendrycks, 2020]
Social reasoning task	ToMBench [Chen, 2024]
Generation: G	60
Population size	10

How does task nature (knowledge / social reasoning) shape evolution?

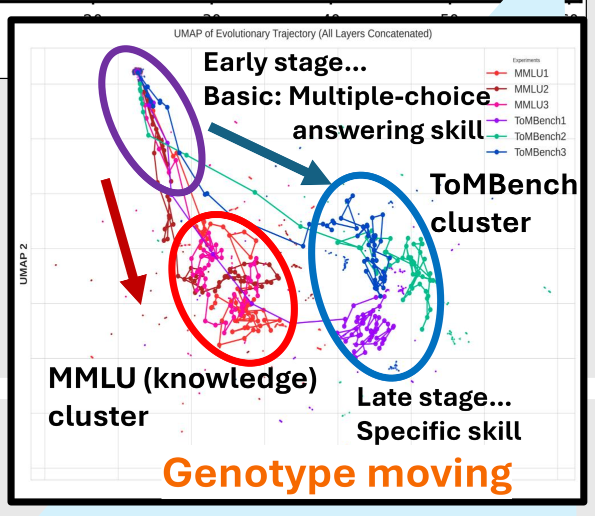
Fitness Trajectories





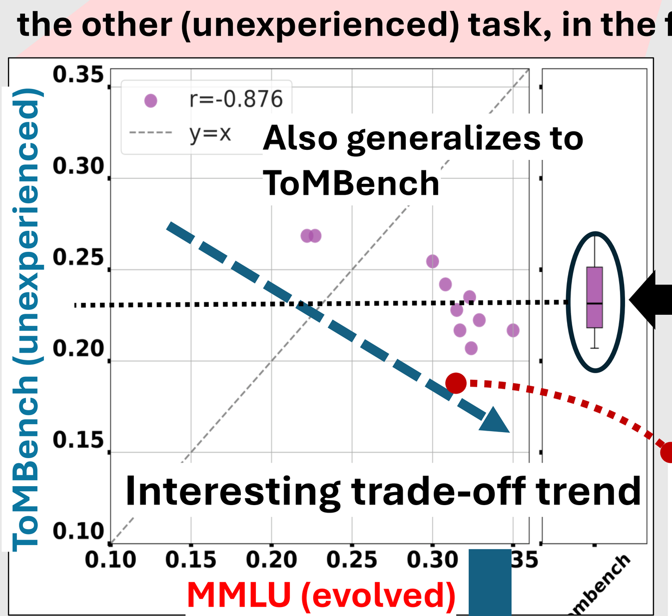
★Successful evolutionary adaptation demonstrated by fitness increase

- Structural differences in fitness landscapes
- A hierarchical process of skill acquisition



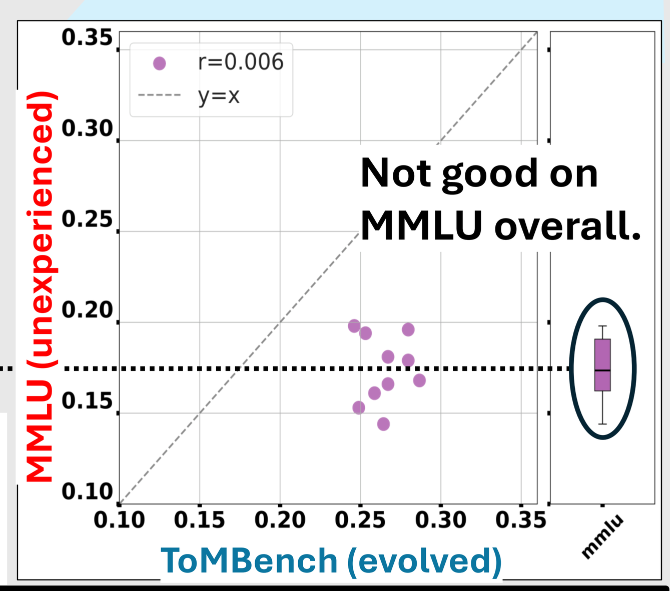
Generalization performance across tasks

Correlation between scores on the evolved task and the other (unexperienced) task, in the final-generation population.

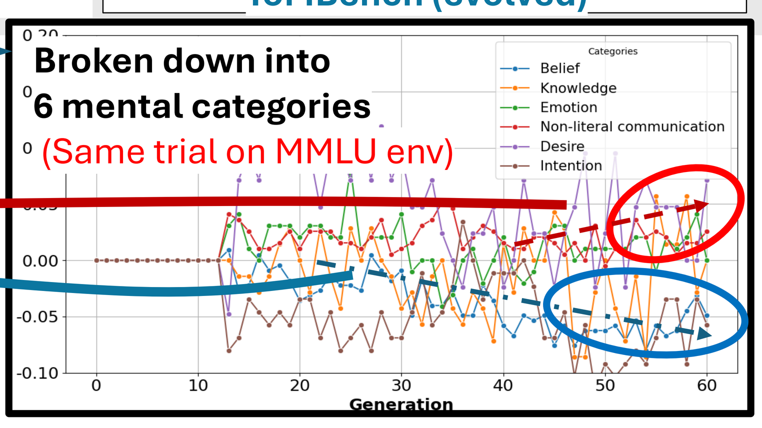


Generalization is asymmetric:
knowledge adaptation carries over to social reasoning,
but not the reverse.

Individuals excelling at MMLU tend to struggle with ToMBench.



Deep dive



Strengths and Weaknesses patterns

Reinforced: Desire, NLC, Emotion
Sacrificed: Belief, Knowledge, Intention

This trade-off is reminiscent of compensatory cognitive development observed in ASD cf. verbal knowledge as a compensatory scaffold for ToM in ASD [Fujino et al., 2015]

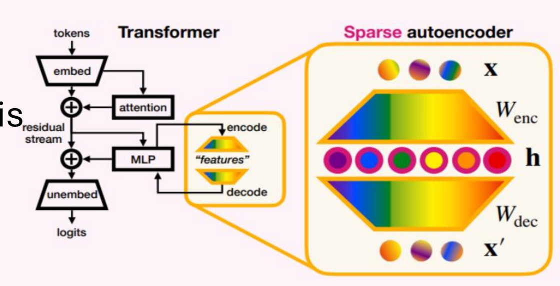
Experiment 2: Internal state analysis of evolved individuals in social reasoning tasks

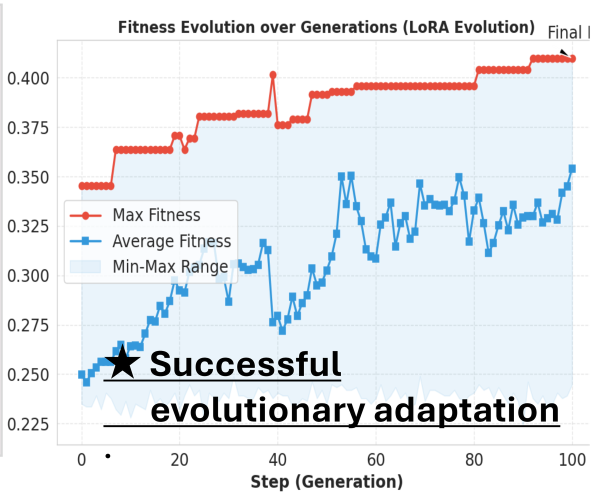
LLM	Llama-3-8b
SAE	LlamaScope lxr 32x [He et al., 2024]
Generation: G	100

Identified a ToM-related SAE feature in the best final-generation individual, then traced its activation back across earlier generations.

Sparse Autoencoder(SAE) [Bricken et al., 2023]

In a Sparse Autoencoder (SAE), the fact that a specific feature exhibits strong activity (strong activation) can be interpreted as a state in which the LLM is strongly recognizing a particular "concept" or "meaning" at that very moment and is thinking (processing information) based on it.





Dynamic, stepwise strategy shift

Early stage (Gen 7, 19):
undifferentiated responses.

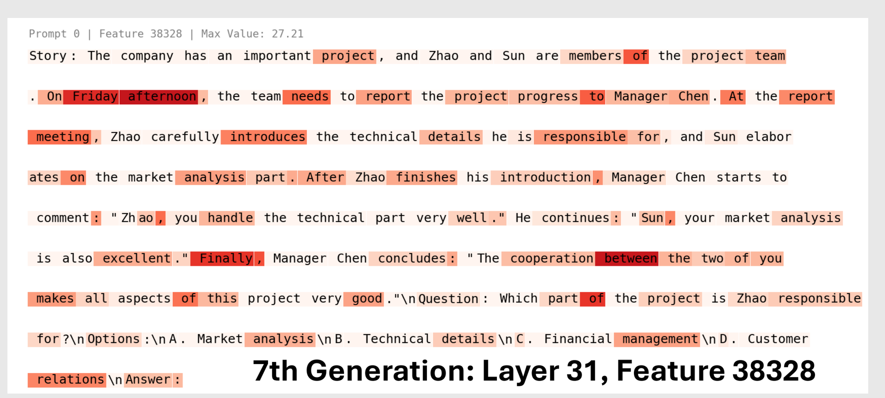
Middle stage (Gen 32, 39):
temporary reliance on superficial cues (e.g., Proper nouns).

Late stage (Gen 81, 92):
specialization in conceptual representations related to ToM.

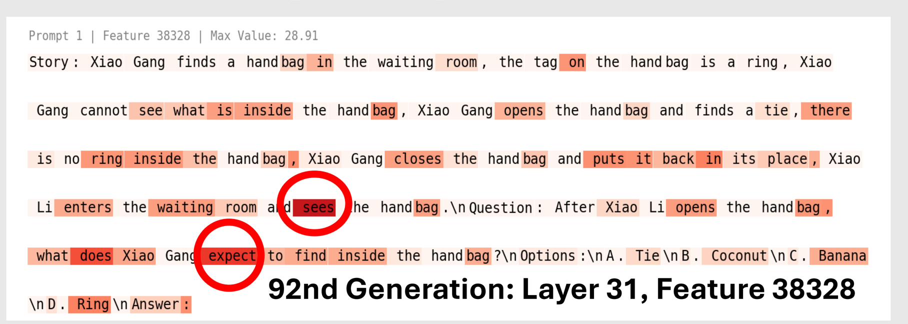
Evolutionary progression of the feature 38328 that evolved as a ToM verb processing circuit

	Base	Gen 7	Gen 19	Gen 24	Gen 32	Gen 39	Gen 81	Gen 92
Proper nouns (Xian, Zhao, etc.)	7.17	4.90	6.86	3.3	11.10	10.58	6.66	6.29
Pronouns (He, his, their, etc.)	5.84	6.04	5.30	5.48	7.24	6.06	4.91	5.13
ToM verbs (thinks, finds, expect, etc.)	15.02	7.67	11.32	11.34	16.95	6.20	13.7	23.94
Other verbs (drive, walk, eat, etc.)	14.22	10.88	7.71	9.92	15.38	11.48	14.3	14.62
Symbols (, . ' " etc.)	7.59	8.80	8.51	7.64	8.15	8.11	8.89	9.54
JSON keyword (option-A, Answer, etc.)	5.42	4.71	3.86	4.28	5.43	4.53	4.84	4.37
Others (, . ' " etc.)	10.13	9.23	8.75	8.68	9.13	8.51	8.67	9.39

Visualization of activation levels for each token regarding specific features.



7th Generation: Layer 31, Feature 38328



92nd Generation: Layer 31, Feature 38328

▲ Activation patterns of Feature 38328 in Gen 7th and 92nd
Darker red indicates stronger activation.

Feature 38328:
Observing the shift in the focus of activation from tokens lacking regular patterns to tokens associated with Theory of Mind (ToM).
Days of the week, spatial relationships, verbs, "of," etc. → "sees," "expect," "thinks," etc.

Conclusion

To elucidate the evolutionary origins of social cognition,
we conducted the adaptive evolution of LLMs as computational model organisms and obtained the following key insights:

Landscape & Divergence:
Task differences shape distinct fitness landscapes, causing evolutionary paths to gradually diverge.

Asymmetric Generalization:
Knowledge adaptation generalizes to social reasoning, but not the reverse (suggesting a cognitive hierarchy).

Internal Focus Shift (SAE):
Representation shifts from superficial cues (proper nouns) to ToM-related verbs, filtering out unnecessary tokens.

Future work

From Spectatorial to Participatory ToM: Transition from static, spectatorial benchmarks to dynamic, multi-agent environments.
An "Evo-Devo" Framework: Integrate evolutionary weight updates (evolution) with online in-context learning (development).

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